

PROFESSIONAL SUMMARY

Senior Computer Science student at Rensselaer Polytechnic Institute, graduating in Fall '26, with a strong interest in large language models (LLMs), AI, and anything technology-based. Seeking an entry-level role where I can build and iterate on LLM- and AI-driven systems while continuing to learn alongside smart engineers.

EDUCATION

- Rensselaer Polytechnic Institute** Troy, NY
Bachelor of Science in Computer Science Aug. 2024 – Dec. 2026
- SUNY Polytechnic Institute** Utica, NY
Bachelor of Science in Computer Science Aug. 2022 – April. 2024

RESEARCH & INDUSTRY EXPERIENCE

- Griffiss Institute** Rome, NY
ATLAS — AI Training Platform (Sole Developer) May 2025 – Present
 - Full-stack AI training platform:** Designed and built ATLAS as sole developer, an LLM-driven scenario training platform pairing an LLM agent with human trainees in a realtime chat environment. Implemented Python agent service on a Redis-backed producer/worker priority queue with multi-turn tool-calling loops, per-room mutual exclusion, and ordered message delivery.
 - Retrieval-augmented memory and MCP tooling:** Built a three-tier hybrid memory system combining structured state files with a Qdrant vector database and sentence-transformers embeddings for semantic retrieval of conversation context. Implemented an MCP server exposing 23 tools, and an LLM-as-a-judge assessment engine using EMA scoring; weighs recent evaluations more heavily to track performance trends over time.
 - Cross-platform dashboard and deployment wizard:** Built a Tauri desktop app (React 19 + TypeScript frontend, Rust backend) with a deployment wizard that transforms into a live instructor dashboard featuring realtime WebSocket message streaming, interactive metrics, a geospatial map panel with track playback, and LLM-powered scenario generation with structured output validation.
 - Containerized infrastructure and performance engineering:** Orchestrated a 6-service Docker Compose stack with host networking. Resolved sustained 100% CPU usage and re-render issues, cutting message latency from 12s to sub-second via WebSocket push and caching.
- Griffiss Institute** Rome, NY
Research Intern – AI/LLM Research May 2025 – Aug 2025
 - Large language model research:** Supported ML R&D on LLMs, designing experiments for training, optimization, and information retrieval. Reviewed LLM/RAG literature and analyzed results for accuracy, latency, and robustness.
 - Tool- and agent-based systems:** Designed tool- and agent-based LLM workflows using MCP-style orchestration and agent-to-agent coordination for multi-step pipelines and external tool interaction, measuring task completion and robustness.
 - Retrieval-Augmented Generation (RAG):** Implemented RAG pipelines spanning ingestion, chunking, embedding, retrieval, and generation; ran experiments on retrieval parameters and prompt strategies to evaluate quality, latency, and correctness trade-offs.
 - Model training with Unsloth:** Fine-tuned LLMs on domain-specific datasets using Unsloth, iterating on configs and hyperparameters with structured experiment sweeps to improve alignment and task performance.
 - HPC and containerized workflows:** Packaged experiments in Docker and deployed them on an HPC cluster, managing jobs, resources, and experiment tracking for large-scale training and evaluation runs.
 - Evaluation and documentation:** Built evaluation harnesses, including LLM-as-a-judge, to evaluate model outputs at scale; documented best practices for future researchers and engineers.
- LiftRyte Start-up** Troy, NY
Co-Founder & Front-End Engineer Jan 2024 – Dec 2025
 - Product development:** Co-founded LiftRyte, a mobile app that helps gym-goers improve exercise form in real time, building front-end components in React Native and shaping the UI.
 - Team collaboration and current status:** Collaborated with a small team on feature design, implementation, and testing; development paused in December 2025 to reflect on future direction.

PROJECTS

- Self-Hosted Homelab** 2026
Personal Arch Linux server – infrastructure, monitoring & DR Jun 2026 – Present
 - Containerized service stack:** Deploy and maintain a Docker Compose stack on Arch Linux, including Gitea, Vaultwarden, and FreshRSS, with Watchtower for automatic updates and Dozzle for log viewing.
 - Network-wide DNS and remote access:** Run AdGuard Home with a local Unbound resolver for network-wide ad-blocking, with remote access scoped entirely through Tailscale.
 - Monitoring, alerting, and backups:** Wrote Bash health-check scripts (disk health, temperature, pending updates, container status) run via cron, with Discord alerts and a healthchecks.io heartbeat; backups handled via restic with tiered retention and least-privilege access controls.
Technologies: Arch Linux, Docker Compose, Tailscale, AdGuard Home, Unbound, restic, Bash, Discord webhooks, healthchecks.io.
- EverQuest Item Finder Bot** 2024
Discord bot for natural-language item lookup
 - NL-to-SQL translation:** Used OpenAI API to translate natural language queries into SQL, designing prompt templates and parsing logic to convert noisy input into structured PostgreSQL queries.
 - Discord bot and web interface:** Exposed item lookup via a Discord bot.
Technologies: Python, Flask, OpenAI API, Discord API, PostgreSQL, HTML/CSS.

SKILLS

- Languages:** Python, Rust, TypeScript, JavaScript, Java, C, C++, HTML & CSS; **ML/AI:** vLLM, ollama, LM Studio, NumPy, HuggingFace Transformers, Unsloth, Qdrant, sentence-transformers; **LLM Tooling:** MCP, RAG, OpenAI function calling, multi-turn tool loops, LLM-as-a-judge evaluation; **Full-Stack:** React, Tauri, Tailwind CSS, Vite; **DevOps/Data:** Docker, Docker Compose, Git/GitHub/GitLab, MongoDB, PostgreSQL, Redis, WebSocket/DDP; **AI Tools:** OpenCode, Claude Code, Hermes